ENSC 427: Communication Networks
Spring 2026 Final Project Presentation

Evaluating QoS Degradation in Online Gaming under UDP Flood using ns3

https://www.sfu.ca/~epa25/

Team 19

John Alanes 301554102 (jpa180@sfu.ca)
Ethan Phan 3014112030 (epa25@sfu.ca)

1. Introduction

This project simulates a DDoS attack in ns3 and evaluates its impact and mitigation on network QoS.

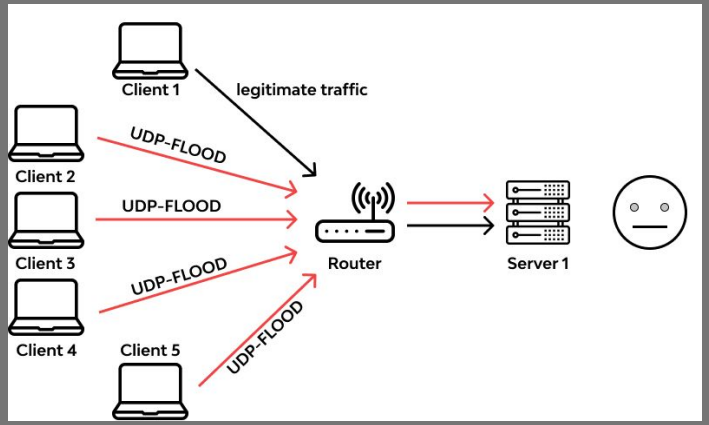
- **Idea**
  UDP flood attack on a UDP server with mitigation (rate limiting + RED)

- **Scope**
  Analyze impact on QoS
  (throughput, delay, packet loss, jitter)

- **Objective**
  Reduce QoS degradation and compare before vs. after mitigation

  Determine an appropriate mitigation technique following existing QoS thresholds

2. Overview of Related Work



What is DDoS

- Distributed Denial-of-Service
- many compromised devices (botnet) send traffic to one target
- make the server unavailable to legitimate users

Why UDP Flood is Effective

- connectionless protocol - no handshake, attackers can send packages instantly
- no congestion control/retransmission limits - sustained high-rate traffic
- server must process each packet - resource exhaustion (CPU, buffers)
- spoofed IPs - harder to trace/block attackers

—

3. Problem Description

UDP Flood overwhelms server → QoS degradation

metrics affected:

- throughput
- delay
- jitter
- packet loss

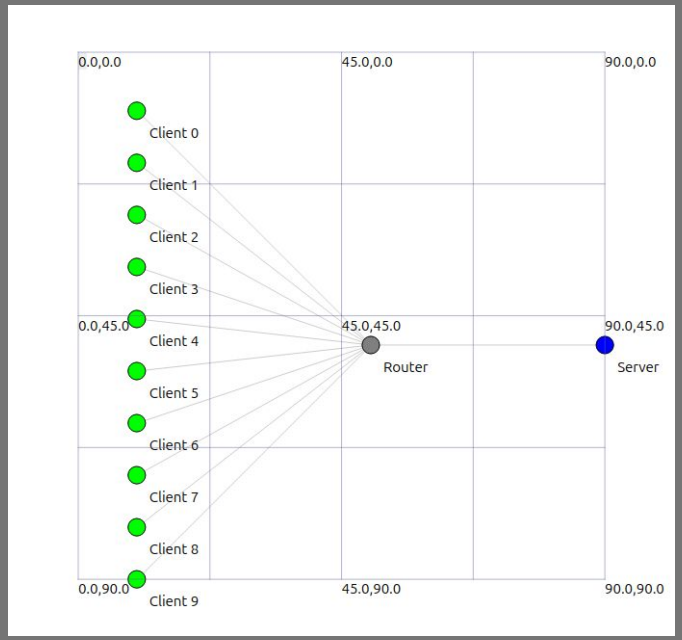
Metric	Acceptable	Degraded	Unacceptable
End-to-End Delay	< 100 ms	100 – 150 ms	**> 150 ms**
Jitter	< 15 ms	15 – 30 ms	**> 30 ms**
Packet Loss	< 1%	1 – 5%	**> 5%**
Throughput Drop	< 30% drop	30 – 70% drop	**> 70% drop**

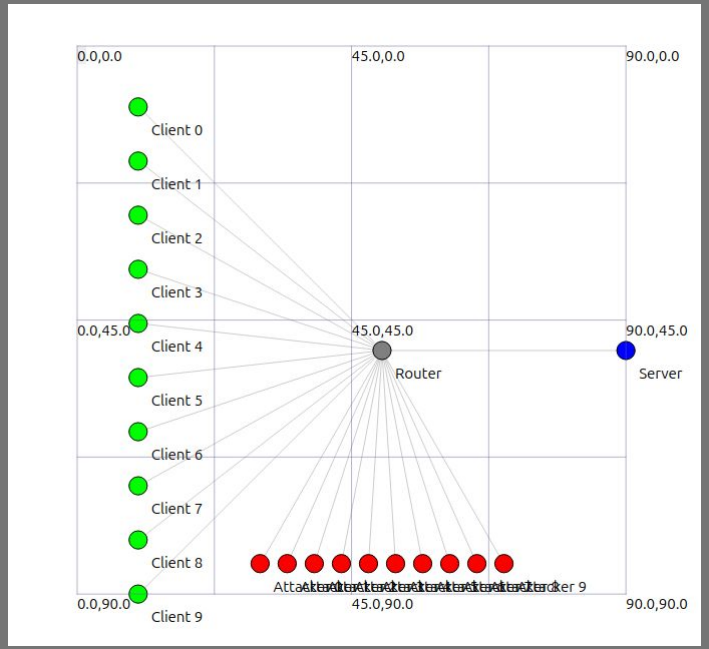
Thresholds derived from Metzger et al. (2022) and Gavrić & Bojović (2022).

4. Implementation

Network Setup: topology diagram (attackers, router, server, users)

NetAnim





4. Implementation

Simulation Parameters

Variable

Number of attackers: 0, 1, 5, 10, 20
Mitigation: None/Blacklist/RED/Both
Attacker data rate: 1MBps

Traffic Settings

Legitimate clients: 128 byte packets every 20ms
Attackers: 1024 byte packets, continuous

Fixed Parameters

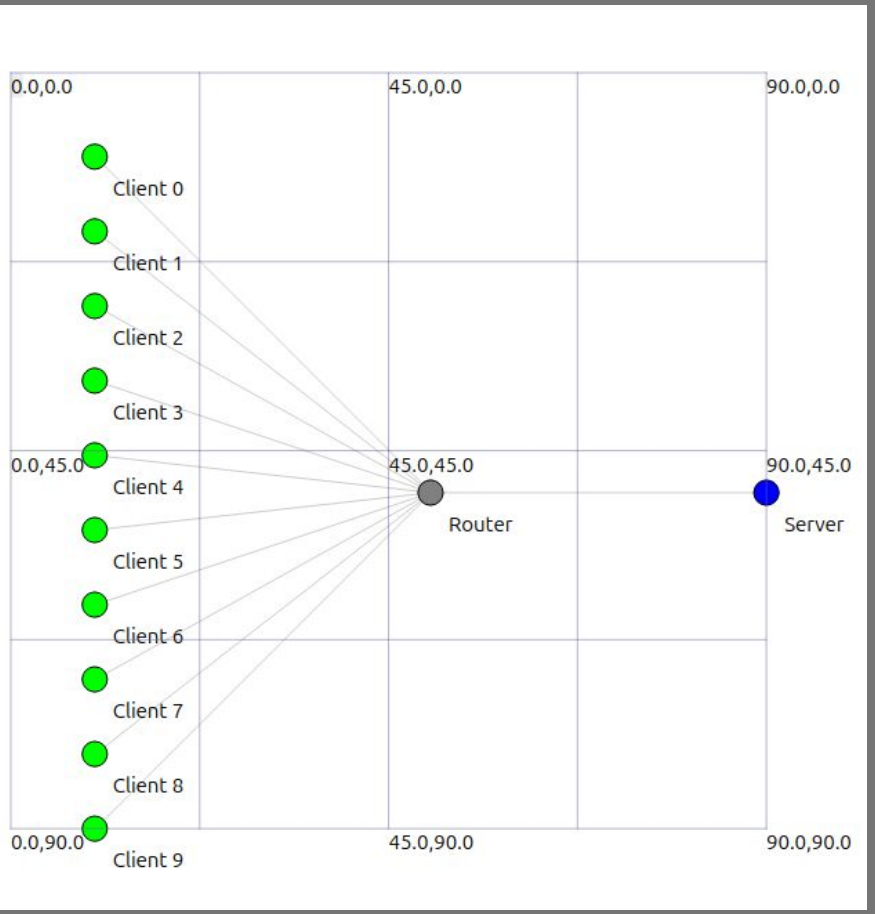
Simulation time 60 sec
Attack start: 5 sec

Network Configuration

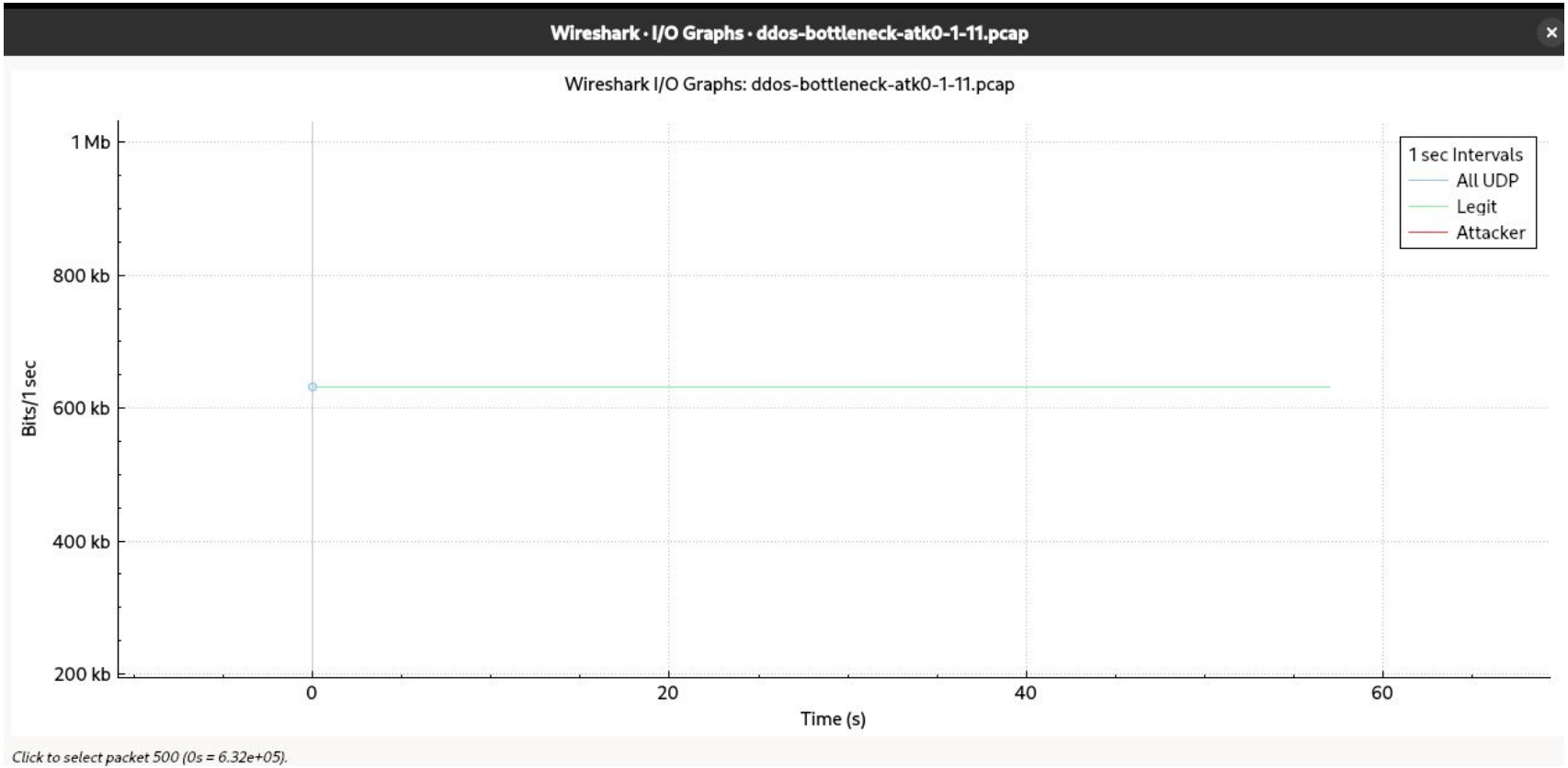
Bottleneck link: 10Mbps, 10ms delay
Access links: 100Mbps, 2ms delay
Protocol: UDP (server port 4000)

4. Implementation Baseline / No Attack: normal QoS

```
int main(int argc, char *argv[])
{
    uint32_t numClients       = 10;
    uint32_t numAttackers     = 0;
    double   simTime          = 60.0;
    bool     blacklistEnabled = true;
```



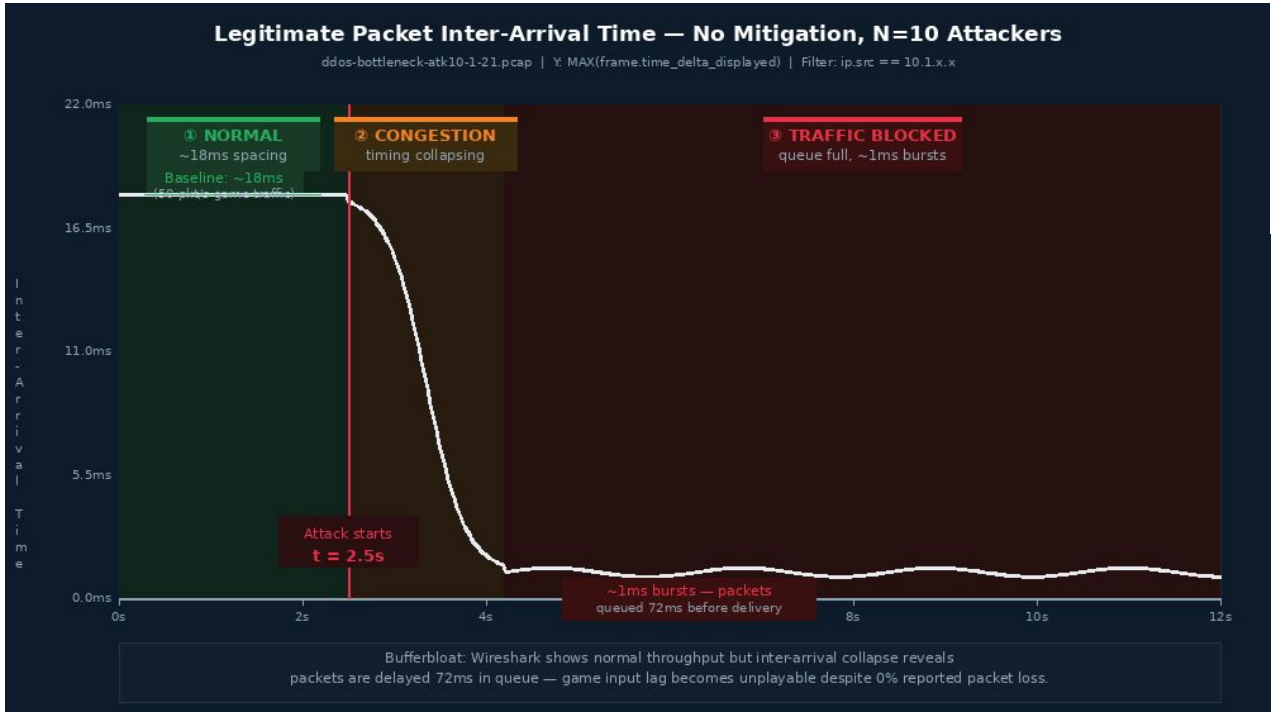
0 Attackers (normal throughput)



4. Implementation
DDoS Attack: QoS Degradation

```
int main(int argc, char *argv[])
{
    uint32_t numClients       = 10;
    uint32_t numAttackers     = 10;
    double   simTime          = 60.0;
```

10 Attackers (DDoS w/ No Mitigation)

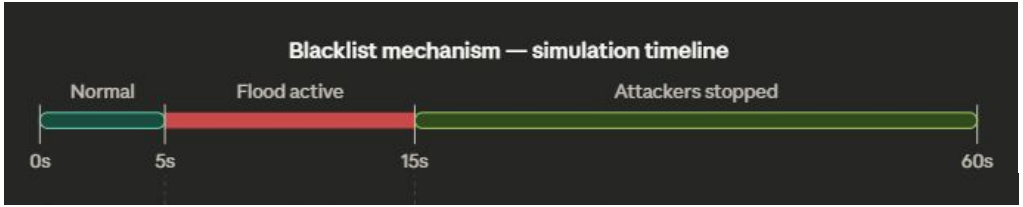


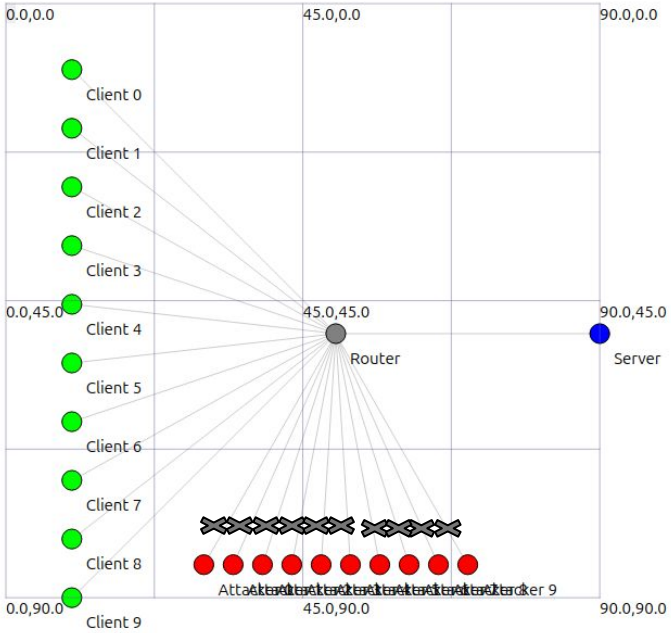
N	Throughput	Delay (ms)	Jitter (ms)
0	0.0624	12.71	0.000
1	0.0624	12.75	0.082
5	0.0624	13.74	2.016
10	0.0624	72.32	0.317
20	0.0624	74.65	4.137

Mitigation Techniques

Blacklisting

- detect when traffic from malicious nodes (traffic thresholding)
- deny all incoming packets from the node

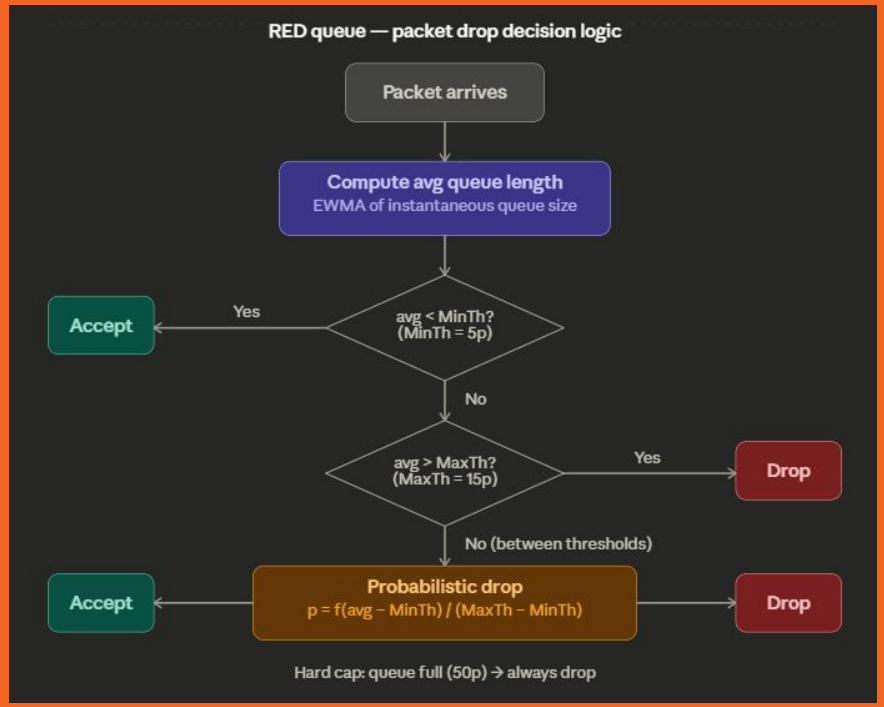




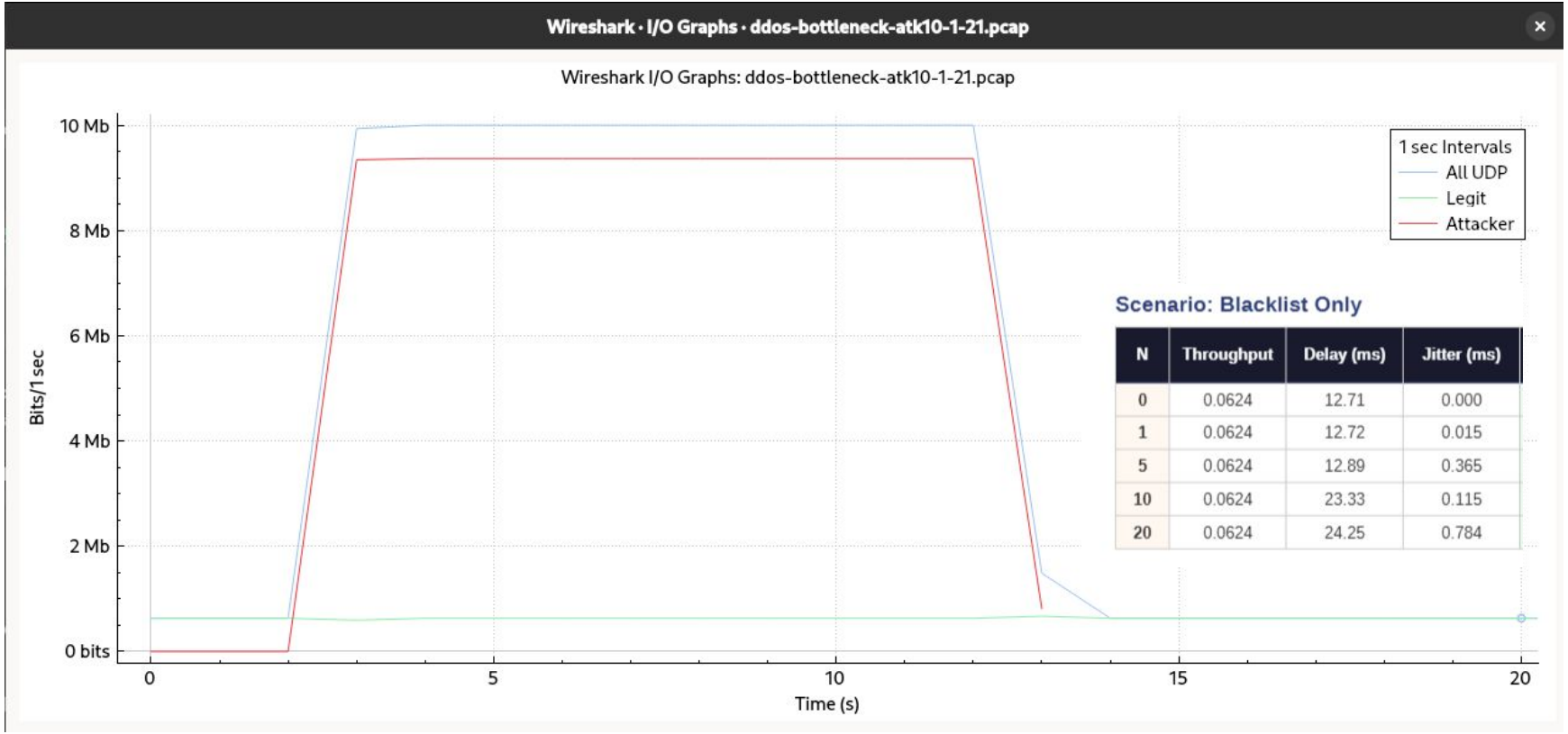
Mitigation Techniques

Random Early Detection (RED) Queuing Algorithm

- RED drops packets **probabilistically** before the queue is full
- MinTh = 5 packets → dropping begins
- MaxTh = 15 packets → all packets dropped
- Prevents queue bloat → controls delay
- More realistic model of real router behaviour under congestion



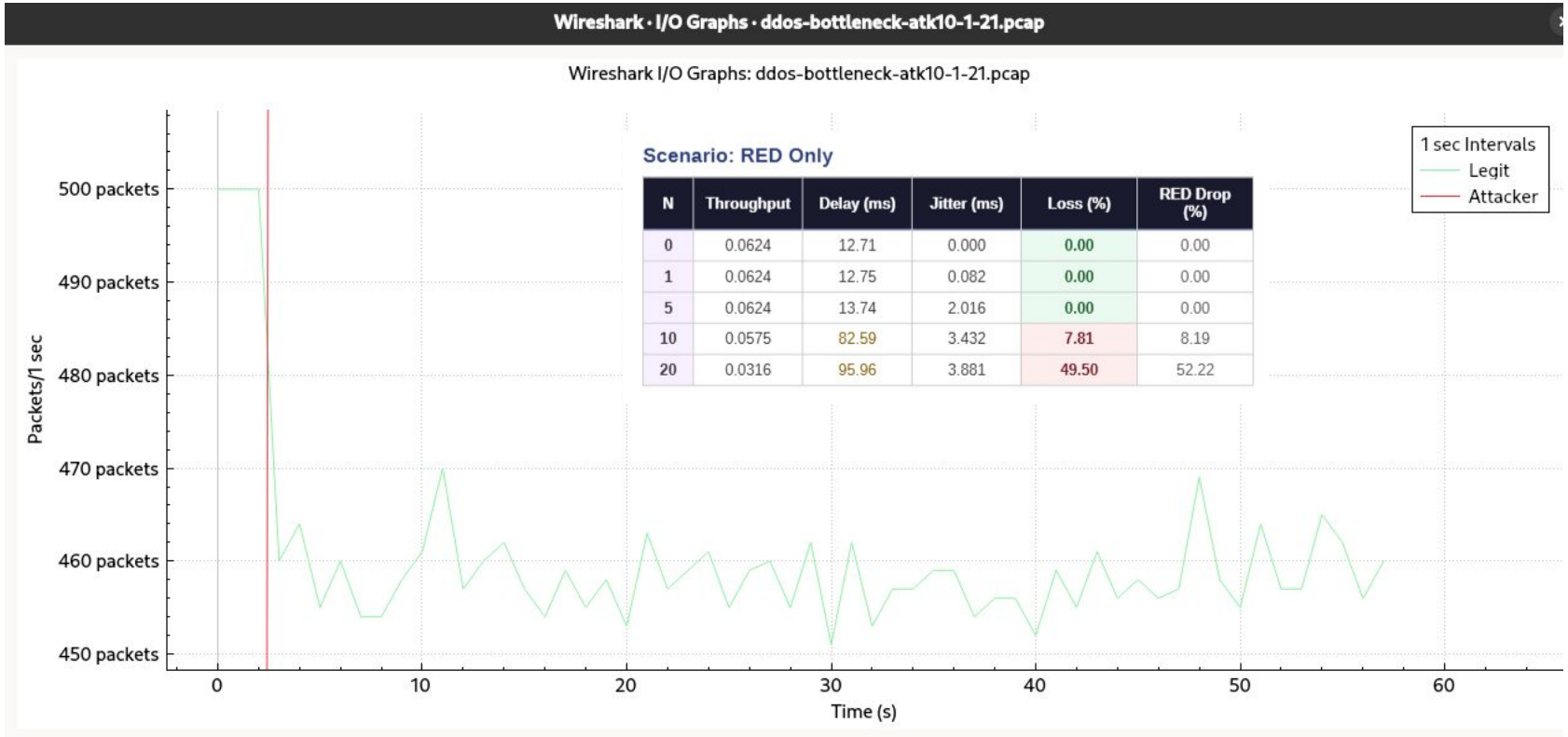
10 Attackers (DDoS w/ Blacklist)



Scenario: Blacklist Only

N	Throughput	Delay (ms)	Jitter (ms)
0	0.0624	12.71	0.000
1	0.0624	12.72	0.015
5	0.0624	12.89	0.365
10	0.0624	23.33	0.115
20	0.0624	24.25	0.784

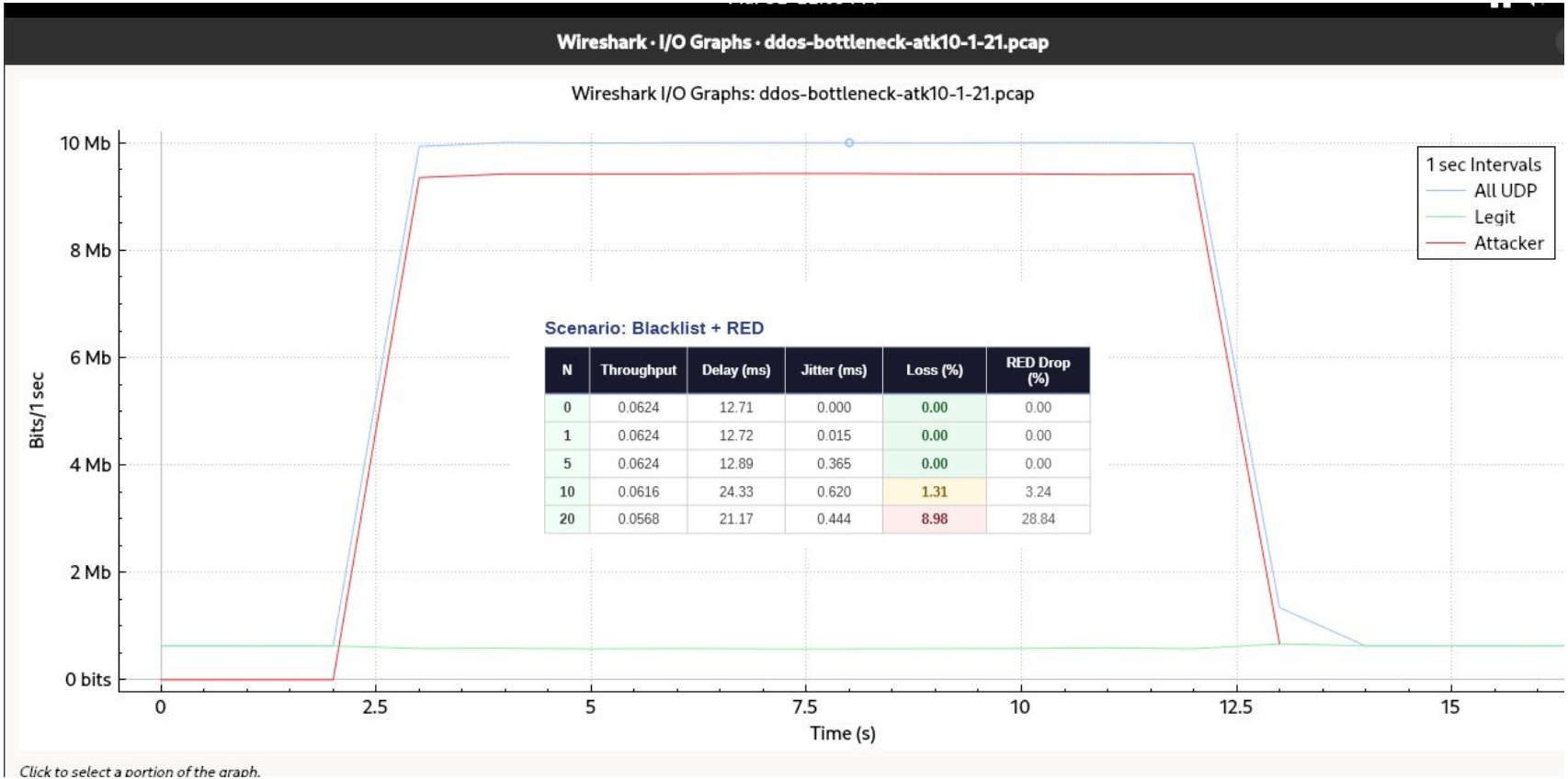
10 Attackers (DDoS w/ RED)



Scenario: RED Only

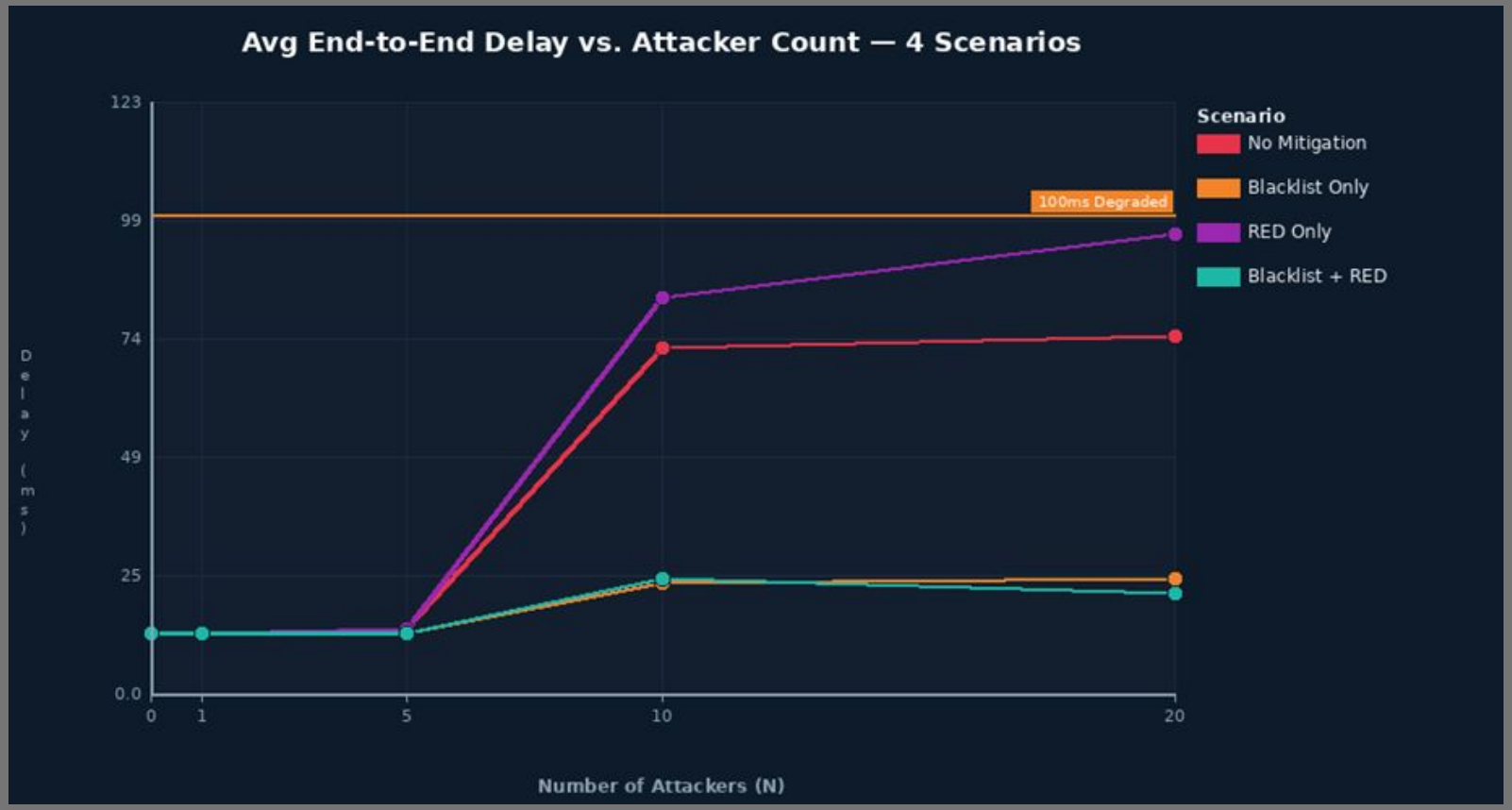
N	Throughput	Delay (ms)	Jitter (ms)	Loss (%)	RED Drop (%)
0	0.0624	12.71	0.000	0.00	0.00
1	0.0624	12.75	0.082	0.00	0.00
5	0.0624	13.74	2.016	0.00	0.00
10	0.0575	82.59	3.432	7.81	8.19
20	0.0316	95.96	3.881	49.50	52.22

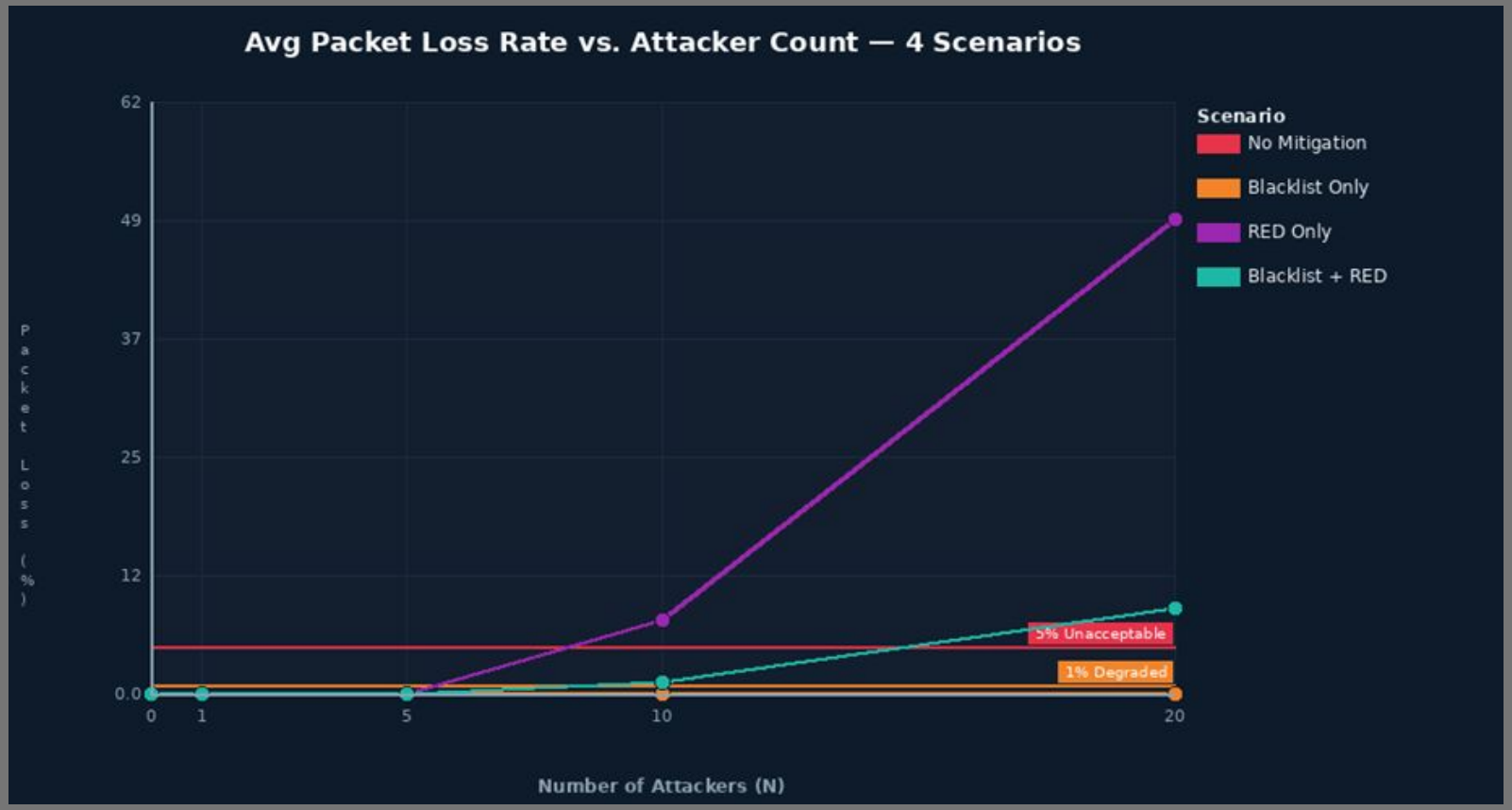
10 Attackers (DDoS w/ Blacklist + RED)



Scenario: Blacklist + RED

N	Throughput	Delay (ms)	Jitter (ms)	Loss (%)	RED Drop (%)
0	0.0624	12.71	0.000	0.00	0.00
1	0.0624	12.72	0.015	0.00	0.00
5	0.0624	12.89	0.365	0.00	0.00
10	0.0616	24.33	0.620	1.31	3.24
20	0.0568	21.17	0.444	8.98	28.84


Avg End-to-End Delay vs. Attacker Count — 4 Scenarios
123
99
74
49
25
0.0
Delay (ms)
100ms Degraded
0
1
5
10
20
Number of Attackers (N)
Scenario
No Mitigation
Blacklist Only
RED Only
Blacklist + RED


Avg Packet Loss Rate vs. Attacker Count — 4 Scenarios
Packet Loss (%)
62
49
37
25
12
0.0
0
1
5
10
20
Number of Attackers (N)
Scenario
No Mitigation
Blacklist Only
RED Only
Blacklist + RED
5% Unacceptable
1% Degraded

5. Discussion

- **Challenges**

  Finding accurate Server Bottleneck value (10-100MBps)

  Adjusting appropriate Attacker data rate and botnet size

- **Improvements/Future Work**

  Develop a better Blacklist algorithm

  Look into more queuing algorithms for comparison

  Learned how to use ns3

6. References

1. D. K. Bhattacharyya and J. K. Kalita, *DDoS Attacks: Evolution, Detection, Prevention, Reaction, and Tolerance*, United Kingdom: Chapman & Hall, 2016.

2. N. Gavrić and Ž. Bojović, “Security Concerns in MMO Games—Analysis of a Potent Application Layer DDoS Threat,” *Sensors (Basel, Switzerland)*, vol. 22, no. 20, p. 7791, Oct. 2022.

3. F. Metzger, S. Geisler, A. Grigorjew, F. Loh, C. Moldovan, M. Seufert, and T. Hosfeld, “An Introduction to Online Video Game QoS and QoE Influencing Factors,” *IEEE Communications Surveys & Tutorials*, vol. 24, no. 3, pp. 1894–1925, Jan. 2022.

4. S. Khozam, G. Blanc, S. Tixeuil, and E. Totel, “QoSentry: A Reinforcement Learning Framework for QoS-Preserving DDoS Mitigation in Software-Defined Networks,” *Journal of Network and Systems Management*, vol. 33, no. 4, Art. 97, Oct. 2025, Springer US.

5. S. Milanov, “A Study on the Impact of DDoS Attacks on the Switching Intent of Online Video Game Players,” University of Twente, Enschede, The Netherlands, 2022.

6. F. Lau, S. H. Rubin, M. H. Smith, and L. Trajkovic, “Distributed Denial of Service Attacks,” Simon Fraser University et al., 2000.

Contributions

Member	Literature Review	ns3 Coding Implementation	Report + Slides
John	50%	50%	50%
Ethan	50%	50%	50%